

AYUSHMAN SAHOO

Software Development Engineer | Machine Learning Engineer

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EDUCATION

B.Tech in Computer Science Engineering (AI & ML) — Aug 2024 – May 2028

VIT-AP University, Amaravati, Andhra Pradesh

TECHNICAL SKILLS

Programming: Python, Java, TypeScript, JavaScript, C, SQL

Software Engineering: Data Structures & Algorithms, OOP, REST APIs, API Integration, Git, GitHub

Frontend: React.js, Next.js, Three.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, Flask

Databases: PostgreSQL, MongoDB

Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning, Reinforcement Learning

AI / Generative AI: LLMs, RAG, AI Agents, Prompt Engineering, LangChain

ML Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

DevOps & Tools: Docker, Linux, Postman, CI/CD

PROJECTS

Telco-RCA 2025 • [GitHub](#)

Python • FastAPI • React • Groq (Llama 3.3 70B) • Docker • Vite • Tailwind CSS

- Built an AI-driven root cause analysis system using graph reasoning and autonomous fault isolation for 5G network outages, deployed live on Hugging Face Spaces.
- Designed a 4-layer hierarchical 5G knowledge graph supporting up to 1,000 nodes across multiple difficulty levels, with causal shuffling and anti-loop safeguards.
- Integrated Llama 3.3 70B through the Groq API for sub-second causal inference using a Functional Cognitive Loop (Trace → Check → Diagnose) grounded on telemetry and alarm feeds.
- Engineered evaluation analytics for MTTR, diagnostic accuracy, and reasoning efficiency; containerized the application with Docker and served it through FastAPI and Uvicorn.

OceanEmbed — Satellite Subsurface Ocean Temperature Reconstruction 2026 • [GitHub](#)

Team Project • Python • PyTorch • FastAPI • TypeScript • React • Docker

- Contributed to a team-built deep learning system for reconstructing 3D subsurface ocean temperature fields from satellite-derived surface ocean variables across the North Indian Ocean.
- Developed and integrated FastAPI REST endpoints for geospatial data processing, model inference, temperature profiles, and ARGO-based validation.
- Implemented the TypeScript/React visualization layer with Vite, API proxying, WebGL heatmaps, and safe-mode fallbacks for reliable demonstrations.
- Containerized frontend and ML backend services using Docker Compose, with persistent data, model, and output volumes for reproducible execution.

Kairo 2025 — Ongoing • [GitHub](#)

React.js • Next.js • TypeScript • Tailwind CSS • Node.js

- Building a full-stack event discovery platform using Next.js, TypeScript, and Node.js to curate tech events, hackathons, concerts, and meetups; deployed on Vercel.
- Designed RESTful APIs and a structured data layer indexing 500+ active events across 50+ cities for event discovery and ranking.
- Implemented authentication, personalized event feeds, and save/bookmark functionality using reusable React components and responsive Tailwind CSS interfaces.

CERTIFICATIONS

- Agentic AI** — Oracle (2026)
- JavaScript Certification** — HackerRank (2026)

- **Java Programming Certification** — CTTC, MSME Affiliated (2025)
- **Python Programming Certification** — CTTC, MSME Affiliated (2025)
- **C Programming Certification** — CTTC, MSME Affiliated (2024)